

Employee Data Management using ServiceNow Import Sets and Transform Maps

Project Report

Date: October 3, 2026

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1. INTRODUCTION

1.1 Project Overview

This project implements an automated employee data management solution on the ServiceNow platform. Employee data prepared in a spreadsheet (.xlsx) is uploaded into a temporary Import Set Table, validated, and moved into a custom target table, Employee Test (u_employee_test), using a Transform Map. Coalesce on Employee ID prevents duplicate records, so repeated imports update existing employees instead of inserting copies. Reports and a dashboard give visibility into the resulting data.

1.2 Purpose

- Replace manual employee data entry with a repeatable bulk import process.
- Maintain accurate, duplicate-free employee records across repeated imports.
- Give administrators and HR teams a single dashboard for employee data insights.

2. IDEATION PHASE

2.1 Problem Statement

In real projects the same employee data is received repeatedly from external sources. Entering it manually is slow and error-prone, and re-importing it can create duplicate or inconsistent records. A reliable, validated, duplicate-safe import process is needed, along with reporting on the imported data.

2.2 Empathy Map Canvas

Quadrant	Description
Says	"I need employee records to be accurate and up to date."
Thinks	Manual uploads risk duplicates and missing fields.
Does	Maintains spreadsheets, uploads files, corrects errors by hand.
Feels	Frustrated by rework; wants confidence in data quality.

2.3 Brainstorming

- Use a custom table (Employee Test) to hold employee records.
- Use Import Sets as a staging area before data reaches the target table.
- Use a Transform Map with Employee ID as the Coalesce field.
- Build reports (by Department, by Location, full list) and combine them into a dashboard.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Prepare: create the spreadsheet with employee data.
2. Upload: load the file via System Import Sets > Load Data.
3. Transform: run the Transform Map to insert or update records.
4. Verify: check the Transform History and the Employee Test table.
5. Analyse: view reports on the Employee Analytics Dashboard.

3.2 Solution Requirement

Functional requirements

- Custom table with fields: Employee ID, Employee Name, Email, Department, Location.
- Import Set Table for temporary staging of uploaded data.
- Transform Map with field mapping and Coalesce on Employee ID.
- Validation of required fields (Employee ID, Name, Email).
- Reports and a dashboard on the Employee Test table.

Non-functional requirements: data integrity, no duplicate records, repeatable imports, role-based sharing of reports.

3.3 Data Flow Diagram

Source	Load Data	Import Set Table	Transform Map + Coalesce	Employee Test table	Reports and Dashboard
Google Sheet / Excel	Upload file	Temporary staging	Validate; insert or update by Employee ID	Final employee records	Employee Analytics Dashboards

3.4 Technology Stack

Component	Technology
Platform	ServiceNow
Source data	Google Sheets, exported as Excel (.xlsx)
Data loading	System Import Sets: Load Data, Import Set Table
Data mapping	Transform Map with Field Maps and Coalesce
Target table	Employee Test (u_employee_test)
Reporting	ServiceNow Reports and Dashboards

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem is duplicate-prone, manual employee data handling. Import Sets with Coalesce solve it directly: data is staged, validated, and then inserted or updated based on Employee ID, keeping the database clean even when the same file is imported repeatedly.

4.2 Proposed Solution

6. Spreadsheet: create a Google Sheet with sample data and download it as Sample Spreadsheet.xlsx.
7. Custom table: create Employee Test (u_employee_test) via Tables > Create New, and add String fields Employee ID, Employee Name, Email, Department and Location using Form Layout.
8. Import Set Table: use Load Data to create and load the staging table (Employee Import) from the spreadsheet (sheet 1, header row 1).
9. Transform Map: name it Sample Spreadsheet Import; source Employee Import; target Employee Test; use Auto Map Matching Fields and save.
10. Transform and validate: run the transform, confirm the success message, and check the records in Employee Test.
11. Coalesce: set Coalesce to true on the Employee ID field map to avoid duplicates.
12. Reports and dashboard: create three reports and add them to the Employee Analytics Dashboards.

4.3 Solution Architecture

The architecture has four layers: Source (spreadsheet), Staging (Import Set Table), Processing (Transform Map with Coalesce and validation), and Target and Presentation (Employee Test table, reports, dashboard). Employee ID is the unique key that decides whether a row is inserted or updated.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Milestone	Activities	Status
1. Spreadsheet and table creation	Create sheet and sample data; create Employee Test table and fields	Done
2. Import Set and Transform Map	Load data; create Transform Map; auto map fields	Done
3. Transform, validate, coalesce	Run transform; verify records; enable Coalesce on Employee ID	Done
4. Reports and dashboard	Create 3 reports; build Employee Analytics Dashboards	Done

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test case	Input	Expected result	Actual result
Initial import	Sample spreadsheet	All rows inserted into Employee Test	Pass: all records created
Re-import with changes	4 rows: 2 existing IDs with changed Name/Email, 2 new IDs	2 inserted, 2 updated	Pass: 2 inserted, 2 updated
Re-import of identical file	Same 4 rows	No inserts or updates	Pass: 0 inserted, 0 updated, 4 ignored

The tests confirm that Coalesce on Employee ID updates changed records, inserts only genuinely new ones, and ignores unchanged data.

7. RESULTS

7.1 Output Screenshots

All records were successfully imported into ServiceNow using the Import Set and Transform Map. The six employee records were created in the Employee Test table, and the dashboard reports reflect the imported data.

The screenshot displays the 'Load Data' configuration page in ServiceNow. The browser address bar shows the URL: `dev443751.service-now.com/now/nav/ui/classic/params/target/create_import_set.do`. The page header includes the ServiceNow logo and navigation links: All, Favorites, History, Workspaces, and a 'Load Data' button. The main form is titled 'Load Data' and contains the following fields:

- Import set table:** Radio buttons for 'Create table' (selected) and 'Existing table'.
- * Label:** Text input field containing 'Employee Import'.
- Name:** Text input field containing 'u_employee_import'.
- Source of the import:** Radio buttons for 'File' (selected) and 'Data source'.
- File:** A 'Choose File' button followed by the text 'sample sheet 23.xlsx'.
- Sheet number:** Text input field containing '1'.
- Header row:** Text input field containing '1'.

A 'Submit' button is located at the bottom of the form. The Windows taskbar at the bottom of the screen shows the time as 9:34 PM on 10/3/2025.

Figure 1: Load Data - creating the Employee Import (u_employee_import) table from the spreadsheet

dev443751.service-now.com/now/nav/ui/classic/params/target/create_import_set.do

servicenow All Favorites History Workspaces Load Data Search

Load Data Submit

Import set table ☐ Create table ☒ Existing table

Import set table Employee Import [u_employee_import]

Source of the import ☒ File ☐ Data source

File Choose File sample sheet 23.xlsx

Sheet number 1

Header row 1

Submit

Figure 2: Load Data - loading new spreadsheet data into the existing Employee Import table

dev443751.service-now.com/now/nav/ui/classic/params/target/u_employee_test_list.do%3Fsysparm_clear_stack%3Dtrue

servicenow All Favorites History Workspaces Employee Tests Search

Employee Tests Department Search Actions on selected rows... New

Employee ID	Employee Name	Department	Email	Location
SB0005	user5	Salesforce	user5@gmail.com	Tamil nadu
SB0006	user6	Salesforce	user6@gmail.com	Tamil nadu
SB0001	user1	Servicenow	user1@gmail.com	Tamil nadu
SB0004	user4	Servicenow	user4@gmail.com	Tamil nadu
SB0003	user3	Servicenow	user3@gmail.com	Tamil nadu
SB0002	user2	Servicenow	user2@gmail.com	Tamil nadu

1 to 6 of 6

Figure 3: Employee Test table showing the imported records (6 of 6)

The screenshot shows the ServiceNow 'Access Controls' page for the 'pa_dashboards' table. The table lists various ACLs with columns for Name, Decision Type, Operation, Type, Active, Updated by, and Updated. The ACLs are as follows:

Name	Decision Type	Operation	Type	Active	Updated by	Updated
pa_dashboards	Allow If	write	record	true	admin	2019-03-25 09:32:03
pa_dashboards	Allow If	write	record	true	admin	2016-04-19 03:29:14
pa_dashboards	Allow If	read	record	true	admin	2019-03-25 09:34:18
pa_dashboards	Allow If	report_view	record	true	admin	2022-11-16 22:19:19
pa_dashboards	Allow If	delete	record	true	admin	2016-04-19 03:26:55
pa_dashboards	Allow If	read	record	true	admin	2018-03-01 07:27:17
pa_dashboards	Allow If	create	record	true	admin	2026-10-02 04:15:38
pa_dashboards	Allow If	delete	record	true	admin	2019-03-25 09:35:29
pa_dashboards.*	Allow If	query_range	record	true	@snc_write_audit@	2026-06-13 04:38:32
pa_dashboards.group	Allow If	write	record	true	admin	2016-07-28 06:29:53
pa_dashboards.notification_link_label	Allow If	query_range	record	true	@snc_write_audit@	2026-06-13 04:38:32
pa_dashboards.notification_text	Allow If	query_range	record	true	@snc_write_audit@	2026-06-13 04:38:32
pa_dashboards.owner	Allow If	write	record	true	admin	2021-11-04 08:32:01

Figure 4: Access Controls (ACL) for pa_dashboards used to enable dashboard creation

The screenshot shows the 'Employee Analytics Dashboards' page with the 'Employee List Report' selected. The report displays a table of employee data with columns for Employee ID, Employee Name, Department, Email, and Location. Below the table, there is a bar chart titled 'Employees by Location' showing the count of employees for each location.

Employee ID	Employee Name	Department	Email	Location
Employee ID: SB0001 (1)				
Employee ID: SB0002 (1)				
Employee ID: SB0003 (1)				
Employee ID: SB0004 (1)				
Employee ID: SB0005 (1)				

Employees by Location

Bar chart showing Employee Count (Y-axis) versus Location (X-axis). The chart displays a single bar for the location 'Employee ID' with a count of 6.

Figure 5: Employee Analytics Dashboard - Employee List Report

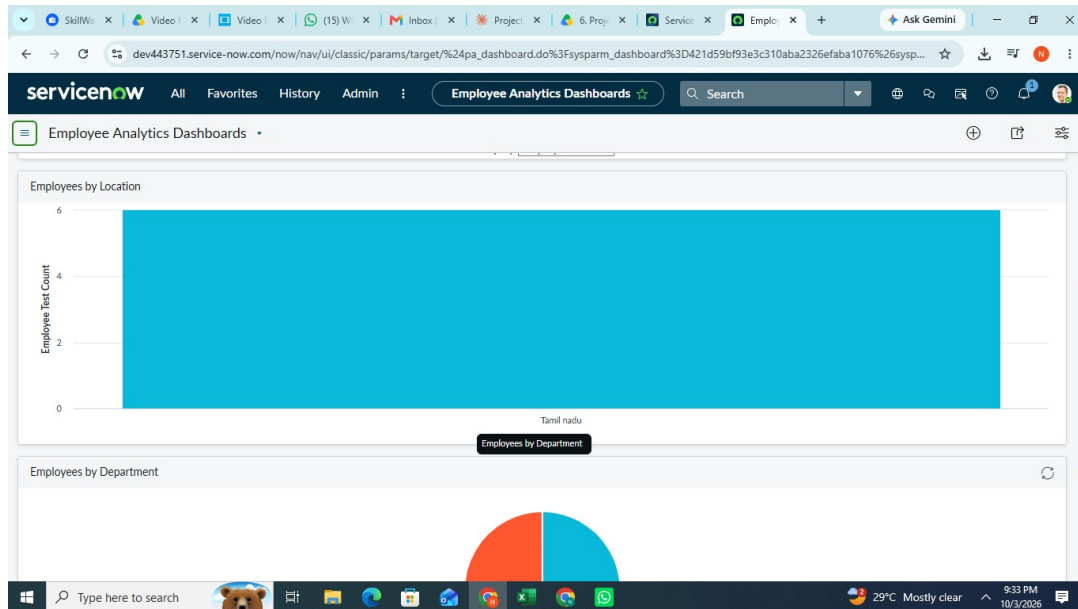


Figure 6: Employee Analytics Dashboard - Employees by Location

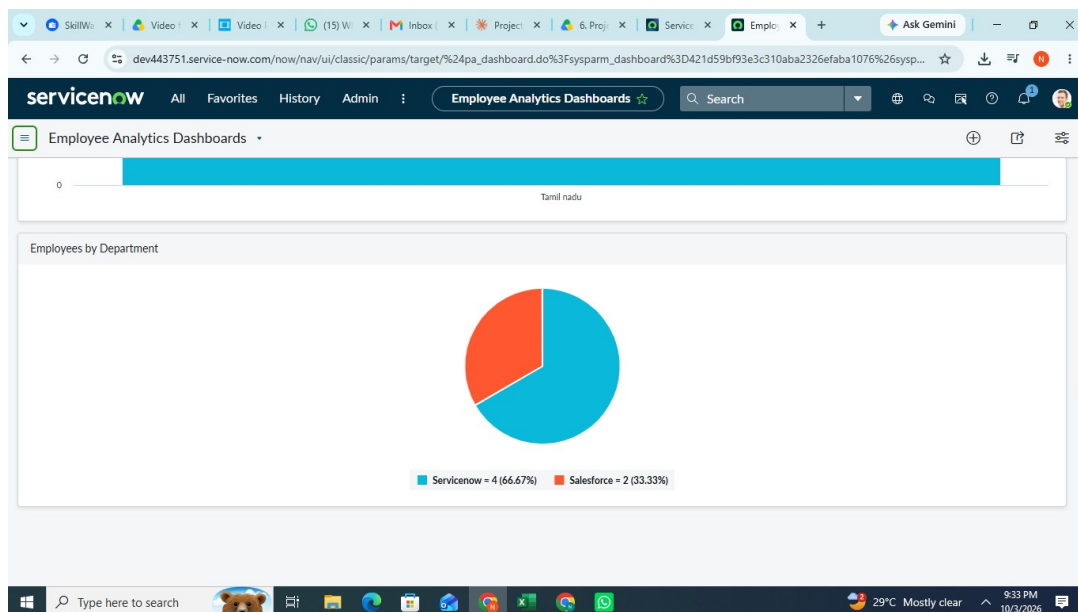


Figure 7: Employee Analytics Dashboard - Employees by Department (Servicenow 4, Salesforce 2)

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages
Prevents duplicate records through Coalesce	Spreadsheet column names must match the mapping
Fast bulk loading instead of manual entry	Coalesce depends on a clean, unique Employee ID

Advantages	Disadvantages
Staging table allows validation before final insert	Needs admin access (e.g., ACL change for dashboard creation)
Dashboard gives real-time visibility	Manual upload step is still required each time
Reports can be restricted by user, group or role	

9. CONCLUSION

This project delivered an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity during employee onboarding and updates. Enabling Coalesce on Employee ID means every import identifies existing employees and updates them rather than inserting redundant entries, which keeps the database clean and consistent.

Reports and a dashboard were added for insight into employee data: department-wise distribution, location-wise distribution and the full employee list. Consolidated on one dashboard, they help administrators and HR teams monitor data trends and quality and reduce manual analysis.

10. FUTURE SCOPE

- Schedule automatic imports (scheduled data imports) from a shared source.
- Add data-quality validation using Transform Map scripts.
- Add reports for import success/error counts, newly added and updated employees.
- Send notifications to HR when an import fails.
- Integrate with HR systems through REST APIs or IntegrationHub.

11. APPENDIX

Source Code: Not applicable (configuration-based, no custom code).

Dataset Link:

[https://velammalitecheduin-my.sharepoint.com/:x/g/personal/cs24058_velammalitech_edu_in/IQCQtWDOjCR-QYZHYA2h6YEuAUPrrfXjle7Subf0vBDIlpE?e=MhzJjO]

GitHub & Project Demo Link: [<https://github.com/Ruthiksha23/Import-Data-using-Transform-Maps-Spreadsheet->] | [https://drive.google.com/file/d/1tvRx-1t_WdkUQLJR8-vC3OB7U6zqgQp5/view?usp=drivesdk]